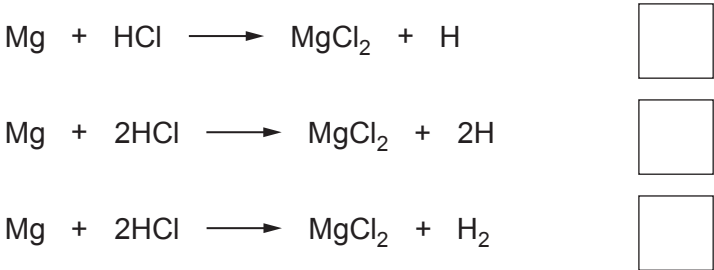
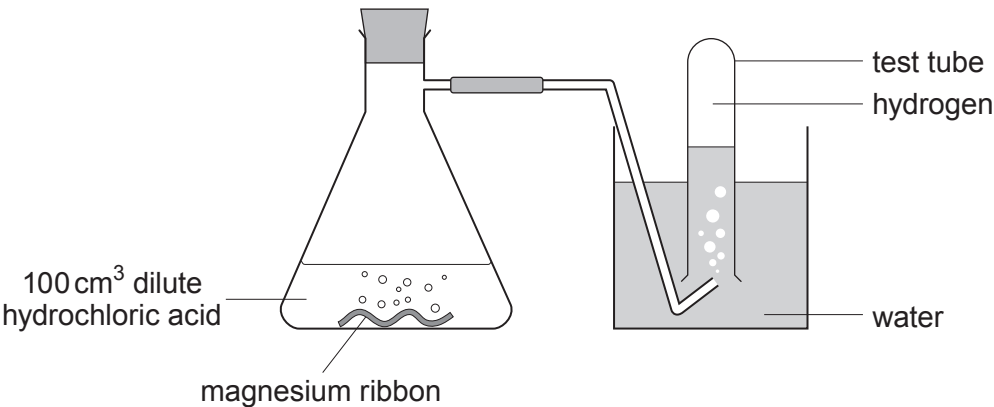


4. Dilute hydrochloric acid reacts with magnesium forming magnesium chloride and hydrogen gas.

(a) Tick (✓) the box next to the correct equation for the reaction between magnesium and hydrochloric acid. [1]



(b) Osian wanted to find out how changing the concentration of the acid affects the rate of the reaction. He carried out five experiments at room temperature (20 °C). He added a 4 cm piece of magnesium ribbon to 100 cm³ of hydrochloric acid of five different concentrations. He recorded the time it took to half-fill a test tube with gas.

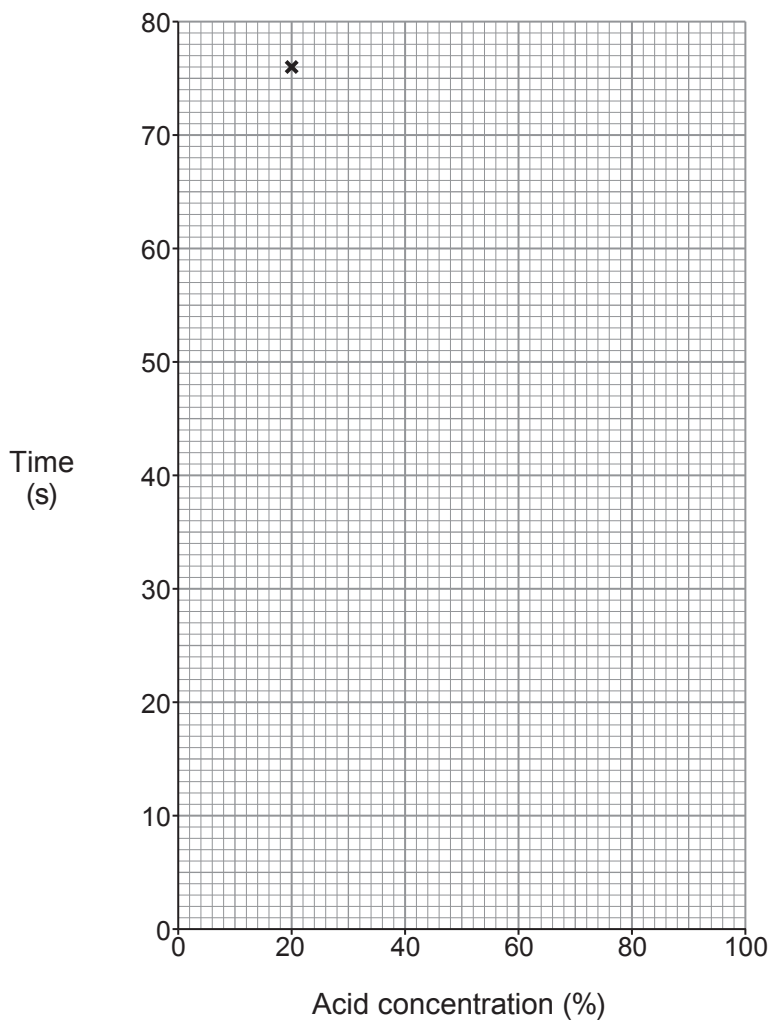


His results are shown below.

Experiment	Acid concentration (%)	Time (s)
1	100	12
2	80	14
3	60	20
4	40	36
5	20	76



- (i) Plot the acid concentration against time on the grid below and draw a suitable line. One point has been plotted for you. [3]



- (ii) Underline the correct word(s) in the brackets to complete the following sentences. [2]

As the acid concentration increases, the **time** to half-fill the test tube with gas
(**increases** / **stays the same** / **decreases**).

As the acid concentration increases, the **rate** of the reaction
(**increases** / **stays the same** / **decreases**).



- (iii) Using your knowledge of particle theory, underline the correct words in the brackets to complete the following sentence. [2]

At a higher concentration, there are (**more** / **less** / **the same number of**) particles present so there will be (**an equal** / **a smaller** / **a greater**) chance of collision.

- (iv) There are other ways the rate of the reaction can be changed.

Tick (✓) the **two** statements that correctly describe other ways the rate of reaction can be increased. [2]

Increasing the temperature of the acid

☐

Using a lump of magnesium

☐

Using a different apparatus

☐

Using magnesium powder

☐

Decreasing the temperature of the acid

☐